

LOGISTICS DATABASE SCHEMA

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1. DRIVERS

- Primary Key: driver_id
- Contains: Driver demographics, employment history, license info

2. TRUCKS

- Primary Key: truck_id
- Contains: Fleet equipment details, acquisition info, status

3. TRAILERS

- Primary Key: trailer_id
- Contains: Trailer inventory, types, status

4. CUSTOMERS

- Primary Key: customer_id
- Contains: Customer accounts, contract types, revenue potential

5. FACILITIES

- Primary Key: facility_id
- Contains: Terminal and warehouse locations, capacity

6. ROUTES

- Primary Key: route_id
- Contains: Origin-destination pairs, distances, rate structures

7. LOADS

- Primary Key: load_id
- Foreign Keys: customer_id, route_id
- Contains: Shipment details, revenue, booking type

8. TRIPS

- Primary Key: trip_id
- Foreign Keys: load_id, driver_id, truck_id, trailer_id
- Contains: Actual trip performance, fuel consumption, duration

9. FUEL_PURCHASES

- Primary Key: fuel_purchase_id
- Foreign Keys: trip_id, truck_id, driver_id
- Contains: Fuel transactions, prices, locations

10. MAINTENANCE_RECORDS

- Primary Key: maintenance_id
- Foreign Keys: truck_id
- Contains: Service history, costs, downtime

11. DELIVERY_EVENTS

- Primary Key: event_id
- Foreign Keys: load_id, trip_id, facility_id
- Contains: Pickup/delivery timestamps, detention, on-time status

12. SAFETY_INCIDENTS

- Primary Key: incident_id
- Foreign Keys: trip_id, truck_id, driver_id
- Contains: Accidents, violations, damage costs

13. DRIVER_MONTHLY_METRICS (Aggregated)

- Composite Key: driver_id, month
- Contains: Monthly performance summaries per driver

14. TRUCK_UTILIZATION_METRICS (Aggregated)

- Composite Key: truck_id, month
- Contains: Monthly equipment utilization summaries

KEY RELATIONSHIPS:

- loads -> customers (many-to-one)
- loads -> routes (many-to-one)
- trips -> loads (one-to-one)
- trips -> drivers (many-to-one)
- trips -> trucks (many-to-one)
- trips -> trailers (many-to-one)
- fuel_purchases -> trips (many-to-one)
- maintenance_records -> trucks (many-to-one)
- delivery_events -> trips (many-to-one)
- safety_incidents -> trips (many-to-one)

ANALYTICAL USE CASES:

1. Driver performance: On-time rates, MPG, revenue per mile
2. Route profitability: Revenue vs costs by lane
3. Fleet utilization: Miles per truck, revenue per asset
4. Maintenance analysis: Cost per mile, downtime impact
5. Fuel efficiency: MPG trends, fuel cost by route
6. Customer analysis: Revenue by customer, service levels
7. Safety metrics: Incident rates, preventable accidents
8. Seasonal patterns: Load volume, rate fluctuations